Healthcare_Energy_Questionnaire

Enumerator's name

-

- ◯ Gian Teufen
- ◯ Till Häussner

Do you give consent to proceed with this survey?

- ◯ Yes
- ◯ No

Questionnaire

» Facility Identifiers

GPS location

You can only proceed once the GPS' accuracy is below 500m.

latitude (x.y °)

longitude (x.y °)

altitude (m)

accuracy (m)

How far is the facility from the distribution lines of the grid? (in km)

Approximation of straight line distance to the nearest point of the grid.

What is the name of the facility?

Facility ID

In the order they are visited.

In which region is the facility located?

- North
- Central
- South

In which district is the facility located?

- Balaka
- Blantyre
- Chikwawa
- Chiradzulu
- Chitipa
- Dedza
- Dowa
- Karonga
- Kasungu
- Likoma
- Lilongwe
- Machinga
- Mangochi
- Mchinji
- Mulanje
- Mwanza
- Mzimba
- Neno
- Nkhata Bay
- Nkhotakota
- Nsanje
- Ntcheu
- Ntchisi
- Phalombe
- Rumphi
- Salima
- Thyolo
- Zomba

What type of facility is this?

- ◯ National referral hospital
- ◯ District/Provincial hospital
- ◯ Health centre/clinic
- ◯ Health post
- ◯ Maternal/child health clinic
- ◯ Dispensary
- ◯ Rural Hospital
- ◯ Other (specify)

What type of facility is this?

_____

What is the managing authority of the facility?

- ◯ Government/Public
- ◯ NGO/Not-for-profit
- ◯ Private-for-profit
- ◯ Mission/faith-based
- ◯ Other (specify)

What is the managing authority of the facility?

_____

Is the facility located in an urban or rural area?

- ◯ Urban
- ◯ Rural

Is the facility outpatient only?

Excluding maternaty ward.

- ◯ Yes
- ◯ No
- ◯ Don't know

Is there a 24/7 maternity ward?

- ◯ Yes
- ◯ No
- ◯ Don't know

What is the catchment population size of the facility?

-99 if not known

_____

What is the catchment area of the facility?

-99 if not known

_____

On average, how many patients visit the facility per day?

- ◯ 0
- ◯ 1-10 patients
- ◯ 11-20 patients
- ◯ 21-50 patients
- ◯ 51-100 patients
- ◯ 101-200 patients
- ◯ >200 patients
- ◯ Don't know

How does the patient population vary over the year? Is there a busy season/which season is the busiest?

- ☐ Hot & Wet Season (November - April)
- ☐ Cool & Dry Season (May - August)
- ☐ Hot & Dry Season (September - October)
- ☐ No significant variation
- ☐ Don't know

Which days of the week is the facility open to patients?

- ☐ Monday
- ☐ Tuesday
- ☐ Wednesday
- ☐ Thursday
- ☐ Friday
- ☐ Saturday
- ☐ Sunday
- ☐ Don't know

Which days are the busiest per week on average?

- ☐ Monday
- ☐ Tuesday
- ☐ Wednesday
- ☐ Thursday
- ☐ Friday
- ☐ Saturday
- ☐ Sunday
- ☐ Don't know

On average, how many hours per day is this facility open?
Excluding off-schedule emergencies and maternity ward

_____

How long do outpatients stay in the clinic on average? (hours)

- ◯ <1 hour
- ◯ 1-2 hours
- ◯ 2-4 hours
- ◯ 4-6 hours
- ◯ >6 hours
- ◯ Don't know

How long is the average consultation?

- ◯ <5 minutes
- ◯ 5-15 minutes
- ◯ 15-30 minutes
- ◯ 30-60 minutes
- ◯ >60 minutes
- ◯ Don't know

How do patients get to the clinic?

- ☐ Foot
- ☐ Bicycle
- ☐ Car (private/taxi)
- ☐ Bus (public transport)
- ☐ Shuttle (facility-provided)
- ☐ Motorbike
- ☐ Oxcart
- ☐ Other (specify)

How do patients get to the clinic?

_____

» » Evening supply

On average, for how many hours during the evening (after 6 p.m.) and night is the facility open for services?
Excluding off-schedule emergencies and maternity ward

- ◯ Never open after 6 p.m
- ◯ Until 8 p.m
- ◯ Until 10 p.m.
- ◯ Until midnight
- ◯ All night long

On average, for how many hours during the evening (after 6 p.m.) and night is electricity available at the health facility?

- ◯ Less than 1 hour
- ◯ 1 to 2 hours
- ◯ 3 to 4 hours
- ◯ 5 to 8 hours
- ◯ 8 to 14 hours

On average, how many nights per month is a patient treated? (after 6 p.m.)

Including off-schedule emergencies, excluding maternity ward.

_____

On average, how many nights (after 6 p.m.) per month is a patient treated without adequate lighting?

_____

» Energy System

Does your facility have electricity from any source (e.g. grid, generator, solar, or other)?

- ◯ Yes
- ◯ No

» » Electricity System

What is the electricity used for in the facility?

- ☐ Stand-alone electric medical devices/appliances (e.g. EPUI cold room, suction apparatus, refrigerator, etc.)
- ☐ Electric lighting (excluding flashlights)
- ☐ Staff housing
- ☐ Communications
- ☐ Device charging
- ☐ Water Pumps
- ☐ Heating, Ventilation, and Air Conditioning (HVAC)
- ☐ Security and Safety (security cameras, alarms, fire detection systems)
- ☐ Other (specify)

What is the electricity used for in the facility?

_____

If the facility has electricity, what is its main source? Note: This means the electricity source that you rely upon most of the time.

- ◯ Central supply (national/community grid)
- ◯ Local mini-grid (i.e., the facility shares electricity supply, typically a renewable source or a generator, with other nearby buildings) Specify the supply modality if known.
- ◯ On-site hybrid solar/generator
- ◯ On-site solar system (except lanterns)
- ◯ On-site generator (with battery storage)
- ◯ On-site generator (no battery storage)
- ◯ Dedicated hydropower system
- ◯ Dedicated wind power system
- ◯ Other (Specify)

If the facility has electricity, what is its main source? Note: This means the electricity source that you rely upon most of the time.

_____

Other than the main or primary source, does the facility have a backup source of electricity?

- ◯ Yes
- ◯ No
- ◯ Don't know

What is the secondary source of electricity?

- ☐ No back-up source
- ☐ Central supply electricity grid
- ☐ Local mini-grid (i.e., the facility shares electricity supply, typically a renewable source or a generator, with other nearby buildings) Specify the supply modality if known.
- ☐ On-site hybrid solar/generator
- ☐ On-site solar system (except lanterns)
- ☐ On-site generator (with battery storage)
- ☐ On-site generator (no battery storage)
- ☐ Dedicated hydropower system
- ☐ Dedicated wind power system
- ☐ Solar lanterns
- ☐ Other (Specify)

Other than the main or primary source, does the facility have a backup source of electricity? If so, what is the secondary source of electricity?

_____

More on this topic will later be recorded in the excel sheet.

_____

Does the facility have an energy storage system like a rechargeable battery system?

- ◯ Yes
- ◯ No
- ◯ Multiple
- ◯ Don't know

If you have both a primary and a secondary/backup source, do you have an energy management system to switch automatically?

- ◯ Yes, automatic shift, with fuel efficiency optimization (for example with the wind/solar)
- ◯ Yes, automatic shift; but no efficiency optimization
- ◯ No, a manual startup is needed

» » Alternative Lighting

Does the facility ever experience complete power loss?

- ◯ Yes
- ◯ No
- ◯ Don't know

If the whole facility has no electricity, what is do you use as a lighting source?

- ☐ No source of lighting
- ☐ Kerosene lamps
- ☐ Candles
- ☐ Battery-operated lamps/flashlights
- ☐ Solar lanterns
- ☐ Headlamps
- ☐ Other (Specify)

If the whole facility has no electricity, what is its backup lighting source?

» » Generator Functionality

Is the generator functional?

- ◯ Yes
- ◯ No
- ◯ Don't know

Does the generator have fuel available today?

- ☐ Yes
- ☐ Petrol
- ☐ Diesel
- ☐ No
- ☐ Don't know

More on this topic will later be recorded in the excel sheet.

» » Solar System Functionality

More on this topic will later be recorded in the excel sheet.

» » Grid Reliability

Is the grid functional?

- ◯ Yes
- ◯ No
- ◯ Don't know

In recent times, how often was there a power outage?

- ◯ Multiple times a day
- ◯ Every day
- ◯ Every 2 days
- ◯ Every 3-4 to days
- ◯ Every week
- ◯ Every two weeks
- ◯ Every month
- ◯ Rarely
- ◯ Never

In the case of a grid power outage, how many hours does it last on average?

-99 if not known

_____

When are supply interruptions most common?

- ◯ Daytime (before dark)
- ◯ Evening (after dark to midnight)
- ◯ Overnight (midnight to 8 a.m.)
- ◯ Don't know

» » Quality

Does the facility usually face a problem of low voltage or voltage fluctuations from the main source?

- ◯ Yes
- ◯ No
- ◯ Don't know

Have voltage fluctuations damaged equipment in the past year?

- No
- Some equipment damaged
- Significant equipment damaged

» » Financial Sustainability

If you use electricity from the grid, did you have sufficient funds available for paying your electricity bills in the last two months?

- Yes
- No
- Don't know
- Not applicable

If you use electricity from the grid, who pays the electricity bill for the health clinic?

- [ ] MoH/DHO with additional budget
- [ ] Donor
- [ ] Health facility baseline budget
- [ ] Health facility comittee
- [ ] Staff for their own housing
- [ ] Patients
- [ ] Don't know
- [ ] Other (Specify)
- [ ] No one

If you use electricity from the grid, who pays the electricity bill for the health clinic?

_____

Over the last two months, did you have sufficient funds available for fuel for the stand-alone generator to operate at all times when it was needed?

- Yes
- No
- Don't know

Who pays for generator fuel?

- [ ] MoH/DHO with additional budget
- [ ] Donor
- [ ] Health facility baseline budget
- [ ] Health facility comittee
- [ ] Staff
- [ ] Patients
- [ ] Don't know
- [ ] Other (Specify)
- [ ] No one

Who pays for generator fuel?

_____

Over the last year, did you have sufficient funds available to keep the solar panels functional?

- ( ) Yes
- ( ) No
- ( ) Don't know

If you use a solar energy-based electricity solution, who pays for light bulbs and solar parts replacement?

- [ ] MoH/DHO with additional budget
- [ ] Donor
- [ ] Health facility baseline budget
- [ ] Health facility comittee
- [ ] Staff
- [ ] Patients
- [ ] Don't know
- [ ] Other (Specify)
- [ ] No one

If you use a solar energy-based electricity solution, who pays for light bulbs and solar parts replacement?

_____

» Origins, Operations and Maintenance

» » Maintenance (preventive)

Is preventive maintenance carried out for any of the energy systems?
Preventive maintenance means the service is carried out even when there is no problem with the system.

- ( ) Routinely
- ( ) Sometimes but not routinely
- ( ) Never

» » Repair (corrective)

Is there a designated staff member trained in managing or troubleshooting any part of the energy system?

- ◯ Yes
- ◯ No
- ◯ Partially
- ◯ Don't know

Elaborate on what the training consisted of.

» » Suitability and Future Needs

Is the current electricity system sufficient for your current needs?

- ◯ Yes
- ◯ No
- ◯ Sometimes
- ◯ Don't know

Do you expect your electricity needs to increase in the next 12 months?

- ◯ Yes
- ◯ No
- ◯ Don't know

Are there plans to upgrade or expand the electricity system?

- ◯ Yes
- ◯ No
- ◯ Don't know

What kind of upgrade is planned?

- ☐ Central supply (national/community grid)
- ☐ Local mini-grid
- ☐ On-site hybrid solar/generator
- ☐ On-site generator (with battery storage)
- ☐ On-site generator (no battery storage)
- ☐ On-site solar system (except lanterns)
- ☐ Dedicated hydropower system
- ☐ Dedicated wind power system
- ☐ Rechargeable battery system
- ☐ Solar lanterns
- ☐ Other (Specify)

What kind of upgrade is planned?

» Climate Change Impact

» » Facility Impact

Have you observed any changes in weather or climate over the past few years?

- [ ] No
- [ ] Increased temperatures
- [ ] Heatwaves
- [ ] More frequent storms
- [ ] Flooding
- [ ] Longer dry periods
- [ ] Other (specify)

Have you observed any changes in weather or climate over the past few years?

Have changes in climate or weather events affected the operations at your facility?

- ( ) Yes, significantly
- ( ) Yes, somewhat
- ( ) No
- ( ) Don't know

What parts of facility operations have been most affected by extreme weather events?

- [ ] Temporary increases in patients due to more malaria in the rainy season
- [ ] Temporary increases in patients due to malnutrition in long dry seasons
- [ ] Temporary increases in patients in cold, dry seasons due to dust in the air, causing cough
- [ ] Temporary decreases in facility accessibility due to floods
- [ ] Temporary decreases in facility accessibility due to wild fires
- [ ] Temporary decreases in facility accessibility due to storms/cyclones
- [ ] Injuries because of storms/cyclones/lightnings
- [ ] Damage to infrastructure or energy systems or equipment because of storms/cyclones/lightnings
- [ ] Grid is less reliable during dry seasons
- [ ] Decreased water availability
- [ ] Migration to/from the region due to weather/environment
- [ ] People not seeking care due to power outages
- [ ] Delayed care-seeking due to transportation issues
- [ ] Reduced ability to refer patients to other facilities due to weather
- [ ] Rooms too hot so doors have to be opened, affects privacy
- [ ] Don't know
- [ ] Other (specify)

What parts of facility operations have been most affected by extreme weather events?

» » Climate Change Perception

Based on your current knowledge, what or who is the cause of climate change?

- ◯ Human activities (e.g. burning fossil fuels, deforestation)
- ◯ Natural processes (e.g. volcanic activity, changes in solar radiation)
- ◯ Both human and natural causes
- ◯ Climate change is not happening
- ◯ I don't know
- ◯ Other (specify)

Based on your current knowledge, what or who is the cause of climate change?

_____

In your opinion, is climate change affecting your community or the way health services are delivered?

- ◯ Yes, significantly
- ◯ Yes, to some extent
- ◯ No
- ◯ Don't know